

ASHWIN SHARMA

✉ ashwinsharma24689@gmail.com

☎ +91 9148678402

🌐 github.com/ashwinsharma24689-ctrl

🎓 Bangalore Institute of Technology

PROFILE

Final-year Electronics and Communication Engineering student with hands-on experience in RTL/digital design, VLSI verification flows, and embedded systems, seeking an internship opportunity to contribute to and grow within the semiconductor industry.

EDUCATION

B.E. in Electronics and Communication Engineering | Bangalore Institute of Technology, Bengaluru | **CGPA: 8.64** | 2023–2027

Relevant Coursework: VLSI Design, Digital Electronics, Microarchitecture, Embedded Systems, Computer Networks.

SKILLS and TOOLS

Category	Skills / Tools
Hardware Description Languages	Verilog
EDA / VLSI Tools	Cadence Genus (Synthesis), Cadence Xcelium (Simulation), Icarus Verilog (Functional Verification), GTKWave (Simulation/Waveform Analysis), Xilinx Vivado (Spartan-3), SEMulator3D
Programming Languages	Embedded Bare Metal C, C Programming, Python
Development Tools & Platforms	Git / GitHub, Arduino IDE, Wokwi Simulator, Keil μ Vision (For ARM Microcontrollers)
Networking & Connectivity	UDP sockets, TCP/IP, WiFi (ESP32), OSI Model, JSON over UDP

PROJECTS

CRISP: Customized RISC-V with Improved Staged Pipeline | Verilog | Microarchitecture |

- Designed a 5-stage pipelined RV32I RISC-V CPU in Verilog with full hazard resolution, EX/MEM and MEM/WB forwarding, load-use stall detection, branch flush, and a 32-bit Carry-Select Adder (CSLA) for reduced critical path
- Implemented complete RV32I ISA support (R/I/S/B/U/J types, all branch variants, byte/halfword/word memory access) with a modular testbench suite covering unit and integration tests across 11 testbenches. [open-sourced](#)

SCORE: Serial Compute and Register Engine | Verilog | RTL Design |

- Designed and implemented a complete 8-bit serial-commanded compute engine in Verilog — UART physical layer, dual synchronous FIFOs, register file, and a CLA-based ALU (10 RISC-V-style opcodes) — targeting 50 MHz FPGA deployment.
- Verified the full design across 11 testbenches (152/152 checks passing) in Icarus Verilog, identifying and fixing 3 functional bugs including a UART reset/FSM race and an incorrect unsigned-comparison flag in the ALU. [open-sourced](#)

Distributed 8-bit Carry-Lookahead Adder | Verilog | Cadence Xcelium/Genus |

- Built and verified an 8-bit CLA adder with a parallel carry-lookahead network to avoid ripple-carry delay; simulated in Xcelium (10/10 test cases passing), synthesized in Genus (TSMC18, 32 cells, 665 μm^2), and validated with RTL-vs-gate-level formal equivalence checking. [open-sourced](#)

WORKSHOPS and CERTIFICATIONS

- SEMulator3D Physical Design Workshop (3 Days):** SEMulator 3D : Learned to build Transistor & FinFET structures along with process variability and device scaling. IISc Bengaluru $\times$ IEEE $\times$ Lam Research | 2025
- RTL-to-GDSII Flow on SRAM (5 Days):** Cadence Xcelium, Genus, Modus, Innovus: Learned to build an S-RAM cell from RTL netlist to verified GDS layout. ECE Dept., BIT $\times$ VLSIMINDS | 2026
- NPTEL – Hardware Modeling Using Verilog** | Prof. Indraneel Sen Gupta, IIT | Course completed | NPTEL | 2025

HACKATHONS & COMPETITIONS

- Cisco Next Gen Champions League 2026:** AI-driven predictive hardware health monitor using edge ML on ESP32/STM32 with multi-sensor fusion | Round 1 cleared
- Xcelerate 24hr Hackathon, NIKSHATRA E-Summit 2025:** MINDCARE+, a student mental health monitoring app | National level, Team Nexus